

SHANG XIANG

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EDUCATION

UCLA ANDERSON SCHOOL OF MANAGEMENT

Los Angeles, CA

Master of Financial Engineering

Dec 2022

- *Relevant Courses:* Stochastic Calculus, Quantitative Asset Management, Data Analytics and Machine learning

NATIONAL UNIVERSITY OF SINGAPORE

Singapore

Bachelor of Science with Honors

June 2018

Major: Applied Mathematics, specialization in Operation Research & Financial Mathematics

Minor: Computer Science, concentration in Machine Learning

SKILLS & CERTIFICATIONS

- *Programming:* Python, R, C++, C#, VBA, Linux Shell Scripting
- *Web Development:* React, Angular, Django, Node.js, PHP
- *Data Processing and Visualization:* SQL, Hadoop, ApacheSpark, MongoDB, Tableau, Pytorch

EXPERIENCE

YAYATI LLC

Los Angeles, CA

Quantitative Developer

April 2023 – Present

- Managing the company's income growth strategy, including weekly collar option strategy; developing option simulation API and user interface under the React framework.
- Developing internal Python API class to optimize investors' portfolios; Leveraging Axioma Python API to build benchmarks and constraints on holdings, as well as objective functions for tax harvesting strategy.
- Conducted stress testing, reverse stress testing and conditional reverse stress testing over portfolios. Researched on portfolios over macro factors composed of ETFs and conducted backtesting over the financial crisis period.
- Built and tested volatility models, e.g., GARCH(1,1) and Negative Momentum to forecast short-term stock market volatility over ETFs on different time horizons.

RUSSELL INVESTMENTS

Seattle, WA

Quantitative Research Analyst Intern

June – September 2022

- Constructed the Python package for systematic trading strategy based on internal equity model, generated output consisting of cycle, valuation and sentiment scores, as well as proposed long-short and dynamic portfolios.
- Upgraded Data ETL pipeline to SQL-based databases and linked with asset model to implement factor analysis.

MIZUHO BANK

Singapore

Associate Data Analyst

January 2019 – July 2021

- Developed business process automation by using Python and PHP under HTTP protocol to set up a direct connection with server to achieve straight-through transactions and send out SWIFT messages to other banks.
- Set up the Tableau server to build interactive dashboards and conducted training sessions for the bank.
- Explored and implemented proof of concept as a full stack developer by using Angular as a frontend framework, Node.js as the backend web server to extract corresponding data from Microsoft SQL server. Presented prototype to headquarters in Tokyo on behalf of the Singapore branch
- Researched customer current accounts and financial profiles, together with the Bloomberg market index to identify salient objectives for machine learning projects. Developed supervised machine learning models to explore potential deposit behaviors; generated early warning indicators to reduce interest rate risk.

PROJECTS

Uncovering the Turnover Dynamics of a Residential Equity Agreement

May – December 2022

- Utilized traditional option pricing methods to build constructive models under an illiquid housing market to replicate the Residential Equity Agreement Turnover Rate and Cash Flow.
- Built machine learning model to predict default rate based on Historical Fund Performance and Real Estate index.

Natural Language Processing, Telegram Chatbot to check real-time stock price

August 2022

- Applied Named-Entity recognition and Intention recognition technology to capture information from human input.
- Deployed the Chatbot on Telegram using Python and extracted real-time stock price by open-source API.

ADDITIONAL INFORMATION

- Singapore Science and Technology "SM2" Scholarship Recipient.